

2024 Statistical Consulting - Final Report

Group 2

2024-05-19

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Data Exploration

Introduction

The laptop dataset is from Kaggle (<https://www.kaggle.com/datasets/siddiquifaiznaeem/laptop-sales-price-prediction-dataset-2024>).

```
knitr::opts_chunk$set(warning = FALSE, message = FALSE)
library(Hmisc, quietly = T)
```

Attaching package: 'Hmisc'

The following objects are masked from 'package:base':

format.pval, units

```
library(tinytex, quietly = T)
```

```
rawdata <- read.csv(
  "C:/github_LTY/Lee_Tsung_Yu/Statistical Consulting/Final Report/data/laptop.csv")
rawdata <- rawdata[, -c(1, 2)]
```

```
latex(describe(rawdata), file = "", caption.placement = "top")
```

27 Variables rawdata
1020 Observations

Brand													
n	missing	distinct											
1020	0	31											
lowest :	Acer	Apple	Asus	ASUS	Avita								
highest:	Ultimus	Walker	Wings	Xiaomi	Zebronics								
Price													
n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95	
1020	0	561	1	82063	59619	25598	31990	43990	63690	94990	149990	199990	
lowest :	8000	12990	13990	14490	14990	highest:	450990	453990	489883	519990	599990		

Rating

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
1020	0	17	0.996	4.374	0.2689	4.00	4.05	4.20	4.35	4.55	4.70	4.75

Value	3.95	4.00	4.05	4.10	4.15	4.20	4.25	4.30	4.35	4.40	4.45	4.50	4.55	4.60
Frequency	1	71	56	69	57	66	68	57	71	76	55	57	64	57
Proportion	0.001	0.070	0.055	0.068	0.056	0.065	0.067	0.056	0.070	0.075	0.054	0.056	0.063	0.056

Value	4.65	4.70	4.75
Frequency	53	64	78
Proportion	0.052	0.063	0.076

For the frequency table, variable is rounded to the nearest 0

Processor_brand

n	missing	distinct
1020	0	7

Value	AMD	Apple	HiSilicon	Intel	MediaTek	Microsoft	Qualcomm
Frequency	250	18	1	742	7	1	1
Proportion	0.245	0.018	0.001	0.727	0.007	0.001	0.001

Processor_name

n	missing	distinct
1020	0	36

lowest :	AMD Athlon	AMD Athlon Pro	AMD Athlon Silver	AMD Ryzen 3	AMD Ryzen 5
highest:	Mediatek	MediaTek	MediaTek Kompanio	Microsoft SQ1	Qualcomm X Elite

Processor_variant

n	missing	distinct
996	24	125

lowest :	1005G1	100U	10210U	10210Y	1035G1
highest:	N5100	N6000	processor	SQ1	Z3735F

Processor_gen

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
891	129	13	0.95	10.45	3.086	5	5	7	12	13	13	13

Value	1	3	4	5	6	7	8	9	10	11	12	13	14
Frequency	1	11	4	92	17	120	7	1	13	113	246	255	11
Proportion	0.001	0.012	0.004	0.103	0.019	0.135	0.008	0.001	0.015	0.127	0.276	0.286	0.012

For the frequency table, variable is rounded to the nearest 0

Core_per_processor

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
1008	12	12	0.977	8.572	4.694	2	4	6	8	10	14	16

Value	2	4	5	6	8	10	11	12	14	16	20	24
Frequency	81	118	8	182	198	189	1	93	69	42	3	24
Proportion	0.080	0.117	0.008	0.181	0.196	0.188	0.001	0.092	0.068	0.042	0.003	0.024

For the frequency table, variable is rounded to the nearest 0

Total_processor

n	missing	distinct	Info	Mean	Gmd
573	447	8	0.902	3.927	2.103

Value	1	2	4	5	6	8	10	12
Frequency	7	221	180	1	121	41	1	1
Proportion	0.012	0.386	0.314	0.002	0.211	0.072	0.002	0.002

For the frequency table, variable is rounded to the nearest 0

Execution_units

n	missing	distinct	Info	Mean	Gmd
573	447	5	0.734	6.998	2.478

Value	4	6	8	12	16
Frequency	193	3	350	3	24
Proportion	0.337	0.005	0.611	0.005	0.042

For the frequency table, variable is rounded to the nearest 0

Low_Power_Cores

n	missing	distinct	Info	Mean	Gmd
1020	0	2	0.124	0.08627	0.1653

Value	0	2
Frequency	976	44
Proportion	0.957	0.043

Energy_Efficient_Units

n	missing	distinct	Info	Sum	Mean	Gmd
1020	0	2	0.124	44	0.04314	0.08263

Threads

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
972	48	14	0.937	12.82	5.988	4	8	8	12	16	20	22

Value	2	4	5	6	8	12	14	16	18	20	22	24	28	32
Frequency	36	50	1	7	178	346	6	218	11	51	27	15	3	23
Proportion	0.037	0.051	0.001	0.007	0.183	0.356	0.006	0.224	0.011	0.052	0.028	0.015	0.003	0.024

For the frequency table, variable is rounded to the nearest 0

RAM_GB

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
1020	0	10	0.798	13.99	6.415	8	8	8	16	16	16	32

Value	2	4	8	12	16	18	32	36	48	64
Frequency	1	25	377	2	543	3	62	1	1	5
Proportion	0.001	0.025	0.370	0.002	0.532	0.003	0.061	0.001	0.001	0.005

For the frequency table, variable is rounded to the nearest 0

RAM_type

n	missing	distinct
998	22	12

Value	DDR3	DDR4	DDR5	DDR6	LPDDR3	LPDDR4	LPDDR4X	LPDDR5	LPDDR5X	LPDDR4X
Frequency	2	522	188	1	1	16	53	173	38	2
Proportion	0.002	0.523	0.188	0.001	0.001	0.016	0.053	0.173	0.038	0.002

Value	PDDR5X	Unified
Frequency	1	1
Proportion	0.001	0.001

Storage_capacity_GB

n	missing	distinct	Info	Mean	Gmd	.05	.10	.25	.50	.75	.90	.95
1020	0	10	0.664	627.7	265	256	512	512	512	512	1000	1000

Value	32	64	128	256	512	1000	1256	1512	2000	4000
Frequency	1	9	19	46	702	218	4	2	18	1
Proportion	0.001	0.009	0.019	0.045	0.688	0.214	0.004	0.002	0.018	0.001

For the frequency table, variable is rounded to the nearest 0

Storage_type

n	missing	distinct
1020	0	4

Value	Hard Disk	NVMe SSD	SSD Hard Disk & SSD
Frequency	14	1	999
Proportion	0.014	0.001	0.979

Graphics_name

n	missing	distinct
1018	2	136

lowest : 10-Core GPU	10 Core GPU	14 Core GPU	18 Core GPU	30 Core GPU
highest: Nvidia RTX 3050	NVIDIA RTX 4050	NVIDIA RTX A500	NVIDIA RTX NVIDIA T550	Radeon Pro 5500M

Graphics_brand

```
      n  missing  distinct
1018      2         7

Value      Adreno      AMD  Apple      ARM  Intel  NVIDIA  Radeon
Frequency      1     151     18      7    486    354      1
Proportion 0.001 0.148 0.018 0.007 0.477 0.348 0.001
```

Graphics_GB

```
      n  missing  distinct    Info  Mean  Gmd
368    652         6    0.871  5.94 2.556

Value      2      4      6      8     12     16
Frequency      4    174     87     80    12     11
Proportion 0.011 0.473 0.236 0.217 0.033 0.030
```

For the frequency table, variable is rounded to the nearest 0

Graphics_integreted

```
      n  missing  distinct
1018      2         2

Value      False  True
Frequency      772  246
Proportion 0.758 0.242
```

Display_size_inches

```
      n  missing  distinct    Info  Mean  Gmd   .05   .10   .25   .50   .75   .90   .95
1020      0         22    0.869  15.16 1.008  13.5  14.0  14.0  15.6  15.6  16.0  16.1

lowest : 11.6 12    12.4 13    13.3, highest: 16.1 16.2 17    17.3 18
```

Horizontal_pixel

```
      n  missing  distinct    Info  Mean  Gmd   .05   .10   .25   .50   .75   .90   .95
1020      0         22    0.569  2036 340.2  1366  1920  1920  1920  1920  2560  2880

lowest : 1080 1200 1280 1366 1440, highest: 3024 3072 3200 3456 3840
```

Vertical_pixel

```
      n  missing  distinct    Info  Mean  Gmd   .05   .10   .25   .50   .75   .90   .95
1020      0         22    0.767  1214 272.7   768  1080  1080  1080  1200  1800  1864

lowest :  768 1024 1080 1200 1280, highest: 2000 2160 2234 2400 2560
```

```

ppi
      n  missing  distinct  Info  Mean  Gmd   .05   .10   .25   .50   .75   .90   .95
1020      0         62  0.902  157.2  30.93  127.3  141.2  141.2  141.2  161.7  212.3  242.6

lowest : 100.45 111.14 111.94 127.34 129.58, highest: 266.26 266.37 267.08 283.02 337.93

```

Touch_screen

```

      n  missing  distinct
1020      0         2

Value      False  True
Frequency    904   116
Proportion 0.886 0.114

```

Operating_system

```

      n  missing  distinct
1020      0         13

lowest : Android 11 OS Chrome OS   DOS 3.0 OS   DOS OS      jio
highest: Mac OS      Prime OS    Ubuntu OS   Windows 10 OS Windows 11 OS

```

Missing data

Next, we explore the missingness of raw data.

```
library(DataExplorer, quietly = T)
plot_missing(rawdata, missing_only = T)
```

